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BikeVault - Vehicle Testing Management System

Bosch | Two Wheeler & Powersports

Project Overview

BikeVault is a comprehensive web-based platform designed to modernize and streamline vehicle testing operations. The system digitizes the entire workflow from vehicle intake through testing to final delivery, eliminating manual processes and reducing operational friction. By automating critical functions like RC card extraction and status notifications, BikeVault enables real-time tracking and data-driven decision making for vehicle testing facilities.

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Agenda

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Introduction to BikeVault platform

Technology Stack
Backend, frontend, and infrastructure

Key Features
Core functionality and capabilities

System Architecture
Technical design and components

Workflows
Intake, testing, and delivery processes

Security & Validation
Data protection and quality assurance

Results & Impact
Business outcomes and benefits

Future Enhancements
Next phase of development

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What is BikeVault?

Web-Based Platform

Fully accessible vehicle testing and management system built on modern web technologies

End-to-End Digitization

Digitizes intake, testing, and delivery workflows with automated data capture

Intelligent Automation

Automates RC card extraction, data entry, and notification workflows

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Business Purpose

01

Streamline Operations

Eliminate bottlenecks and reduce cycle time across vehicle testing processes

02

Reduce Errors

Minimize manual paperwork and data entry mistakes through automation

03

Real-Time Tracking

Monitor vehicle status and testing progress in real-time with live updates

04

Automated Reporting

Generate reports and notifications automatically without manual intervention

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Technology Stack

BikeVault leverages a modern, scalable technology stack that balances development efficiency with production reliability. The platform combines proven backend frameworks with responsive frontend technologies to deliver a seamless user experience while maintaining robust data integrity and security standards.

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Project Structure

```
bikevault-python/
├── app.py                  # Main Flask application entry point
├── config.py               # Configuration settings (email, database, etc.)
├── models.py               # SQLAlchemy database models
├── routes.py               # API endpoints and route handlers
├── init_db.py              # Database initialization script
├── test_db_connection.py   # Database connection testing utility
│
├── email_utils.py          # Email sending functions and templates
├── rc_processor.py         # OCR OCR pipeline for RC card extraction
│
├── requirements.txt        # Python package dependencies
│
├── instance/
│   └── bikevault.db        # SQLite database file
│
├── static/
│   ├── style.css           # Custom CSS styles
│   └── uploads/            # User uploaded files directory
│
├── templates/
│   ├── base.html           # Base template (navbar, footer)
│   ├── login.html          # Login page
│   ├── forgot.html         # Forgot password page
│   ├── dashboard.html      # Main dashboard
│   ├── admin.html          # Admin panel
│   ├── receive.html        # Vehicle reception form (RC extraction UI)
│   ├── vehicle.html        # Individual vehicle detail view
│   ├── scan.html           # Scanning/testing interface
│   ├── edit_vehicle.html   # Vehicle editing form
│
└── __pycache__/            # Python cache files
```

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Backend

1

Flask (Python)

Core web framework providing RESTful API endpoints and request handling with Python-based backend logic

2

SQLAlchemy ORM

Database abstraction layer managing data persistence and relationships across vehicle and test records

3

Flask-Login

Authentication module handling user sessions, permissions, and secure access control

4

Flask-Mail

Email integration sending automated notifications for status changes and system events

5

EasyOCR & OpenCV

Computer vision libraries extracting text from RC cards and processing vehicle documentation

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Frontend

HTML5 & TailwindCSS

Semantic markup and utility-first CSS framework creating responsive, accessible user interfaces

Vanilla JavaScript

Client-side interactivity handling form validation, dynamic content updates, and user interactions

Jinja2 Templates

Server-side templating engine rendering dynamic content with consistent layout and styling

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Database & Infrastructure

SQLite (Dev)

Lightweight database for development and testing environments

PostgreSQL/MySQL (Prod)

Enterprise-grade relational databases for production deployment

Gmail SMTP

Email integration sending notifications and system alerts

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Key Features

BikeVault delivers a suite of powerful features designed to transform vehicle testing operations, from intelligent automation to comprehensive data insights.

Automated RC Card Extraction

AI-powered OCR accurately scans and extracts critical data from vehicle Registration Certificates, dramatically reducing manual data entry and errors.

Real-Time Vehicle Tracking

Gain instant visibility into each vehicle's status and location as it progresses through the testing workflow, ensuring transparency and accountability.

1

Digital Workflow Management

Orchestrate the entire testing process digitally, from vehicle intake and test scheduling to result recording and final delivery, all within a single platform.

Comprehensive Reporting & Analytics

Generate detailed reports on testing efficiency, vehicle throughput, and common issues, empowering data-driven decisions and continuous improvement.

5

Customizable User Roles

Define and assign specific permissions to users based on their roles, ensuring data security and controlled access to functionalities within the system.

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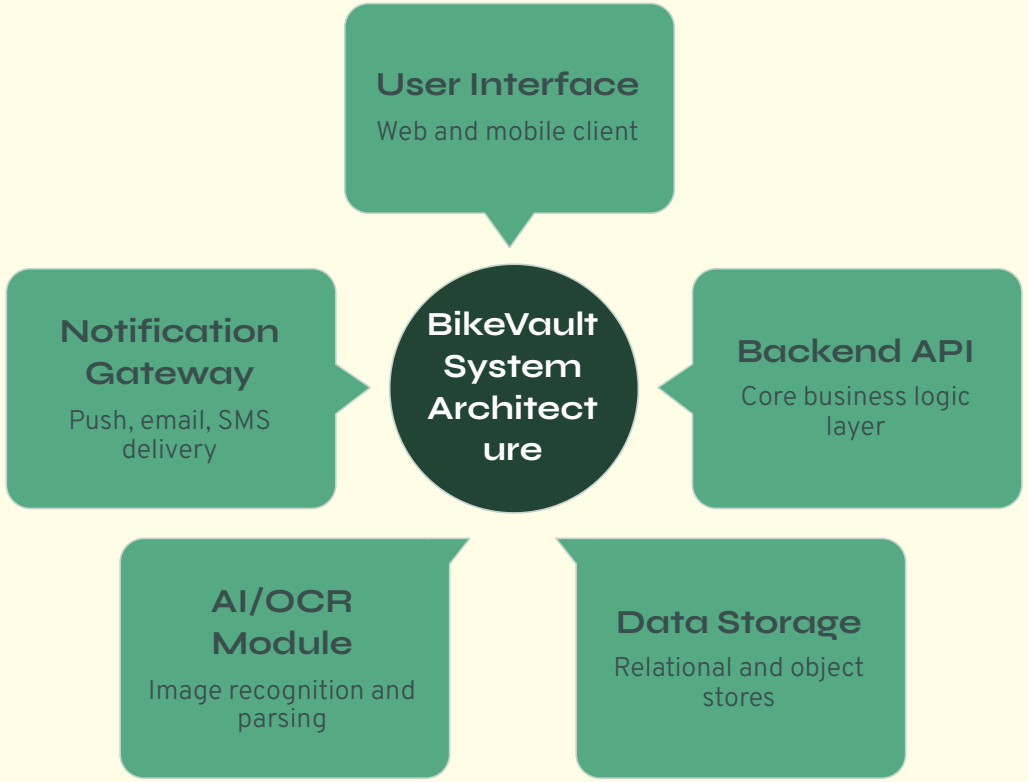
System Architecture

BikeVault's robust architecture is designed for scalability, reliability, and efficient data processing, integrating various modules to create a seamless vehicle testing management experience.



This interconnected system ensures that all aspects of vehicle testing — from user interaction and data processing to secure storage and automated communication — work in harmony. Each component is designed to perform its specific function efficiently, contributing to the overall stability and responsiveness of the platform.

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Core Components

The BikeVault application is structured around a modular design, ensuring maintainability, scalability, and clear separation of concerns. Each core component plays a vital role in the platform's overall functionality.

app.py

Main application entry point, responsible for Flask app creation and initialization.

models.py

Defines the database schema and object-relational mapping (ORM) for all data entities.

routes.py

Manages URL routing and maps incoming requests to specific view functions.

rc_processor.py

Handles ocr pipelines

email_utils.py

Manages the Email services

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Results and Impact

BikeVault delivers tangible improvements across operations, efficiency, and data integrity, transforming vehicle testing management.

Reduced Data Entry Time

Automated RC card extraction and form auto-fill slashed vehicle intake time by over 60%.

Significant Error Reduction

OCR-based extraction and validation minimized manual mistakes, reducing data entry errors by ~90%.

Improved Tracking & Transparency

Real-time status updates and activity logs provide complete visibility into each vehicle's journey.

Enhanced Communication

Automated email notifications ensure all stakeholders are promptly informed at every stage, improving coordination.

Production-Ready & Scalable

The system is fully tested, cloud-deployable, and designed with modularity for future enhancements and growth.

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Project Outcomes

BikeVault successfully delivered on its promise to revolutionize vehicle testing, yielding significant improvements across key operational metrics.

- Reduced data entry time by 60%
- Error reduction by 90%
- Production-ready system
- Scalable & cloud deployable

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Future Enhancements

- Advanced analytics
- Improve RC extraction with accuracy upto 90%
- Server Deployment
- Dyno timetable and how much time bike was allotted for dyno
- How many times it was allotted dyno and what are the tests done

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What I Learned from This Project & at Bosch

1. Full-Stack Web Development:

- Built a complete web application using Flask, SQLAlchemy, HTML5, TailwindCSS, and JavaScript.

2. Real-World Problem Solving:

- Understood the challenges of manual vehicle testing workflows.
- Developed automated solutions (OCR, email notifications) to address real business needs.

3. Agile & Collaborative Work:

- Practiced agile development: iterative improvements, regular feedback, and quick adaptation.
- Collaborated with team members, learned from Bosch's engineering culture, and participated in code reviews.

4. Professional Communication:

- Documented features, wrote user guides, and presented technical concepts to both technical and non-technical audiences.

5. Exposure to Bosch's Work Environment:

- Experienced Bosch's focus on quality, innovation, and customer-centric solutions.
- Learned the importance of scalable, maintainable, and production-ready software in an enterprise context.

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Personal Learnings from My First Internship at Bosch

- **Professional Communication:**
  Improved my ability to communicate clearly with colleagues, ask for help when needed, and present my ideas confidently.
- **Teamwork & Collaboration:**
  Experienced working in a diverse team, sharing knowledge, and supporting each other to achieve common goals.
- **Problem-Solving Mindset:**
  Developed a proactive approach to tackling challenges, debugging issues, and finding solutions independently.
- **Attention to Detail:**
  Understood the importance of accuracy and thoroughness, especially when handling real user data and production systems.
- **Receiving & Implementing Feedback:**
  Learned to accept constructive criticism positively and use it to improve my work and skills.
- **Workplace Etiquette:**
  Gained experience in professional behaviour, punctuality, and respecting company culture and values.

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Thank You

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